

Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects*

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Abstract

To estimate the dynamic effects of an absorbing treatment, researchers often use two-way fixed effects regressions that include leads and lags of the treatment. We show that in settings with variation in treatment timing across units, the coefficient on a given lead or lag can be contaminated by effects from other periods, and apparent pretrends can arise solely from treatment effects heterogeneity. We propose an alternative estimator that is free of contamination, and illustrate the relative shortcomings of two-way fixed effects regressions with leads and lags through an empirical application.

Keywords: difference-in-differences, two-way fixed effects, pretrend test

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following differences. First, in the setting where there is no never-treated group, our method uses the last cohort to be treated as a control group, whereas Callaway and Sant’Anna (2020a) use the set of not-yet-treated cohorts. Our method and theirs thus rely on different, but non-nested parallel trends assumptions. Second, our estimation method can be cast as a regression specification and thus may be more familiar to applied researchers. However, a third difference is that the procedure of Callaway and Sant’Anna (2020a) allows for conditioning on time-varying covariates. de Chaisemartin and D’Haultfœuille (2020) and Goodman-Bacon (2018) respectively propose alternative estimators and diagnostic tools for estimation of causal effects in staggered settings, but do not consider the estimation of the dynamic path of treatment effects as we do.

2 Event studies design

In this section we first formalize the “event studies design”. As discussed in Section 2.3, based on how this term is deployed in the empirical literature, an *event study* design is a *staggered adoption* design where units are treated at different times, and there may or may not be never treated units. It also nests a *difference-in-differences* design, where units are either first treated at time t_0 or never treated.

Specifically, we consider a setting with a random sample of N units observed over $T + 1$ time periods, where T is fixed. For each $i \in \{0, \dots, N\}$ and $t \in \{0, \dots, T\}$, we observe the outcome $Y_{i,t}$ and treatment status $D_{i,t} \in \{0, 1\}$: $D_{i,t} = 1$ if i is treated in period t and $D_{i,t} = 0$ if i is not treated in period t . Throughout we assume that the observations $\{Y_{i,t}, D_{i,t}\}_{t=0}^T$ are independent and identically distributed (i.i.d.).

In the general case of event studies we focus on an *absorbing* treatment such that the treatment status over time is a non-decreasing sequence of zeros and then ones, i.e. $D_{i,s} \leq D_{i,t}$ for $s < t$. We can thus uniquely characterize a treatment path by the time period of the initial treatment, denoted with $E_i = \min \{t : D_{i,t} = 1\}$. If unit i is never treated i.e. $D_{i,t} = 0$ for all t , we set $E_i = \infty$. Based on when they first receive the treatment, we can also uniquely categorize units into disjoint cohorts e for $e \in \{0, \dots, T, \infty\}$, where units in cohort e are first treated at the same time $\{i : E_i = e\}$.

We define $Y_{i,t}^e$ to be the potential outcome in period t when unit i is first treated in time period e . We define $Y_{i,t}^\infty$ to be the potential outcome if unit i never receives the treatment, which we call the “baseline outcome”. Since the timing of the initial treatment uniquely characterizes one’s treatment path, we can

the closed-form expressions for the weights using the classical “omitted variables bias formula” in Section 3.6. We defer the formal derivation to the Appendix. Here we mention the following four properties of the weights $\omega_{e,\ell}^g$.

- For relative periods of μ_g ’s own bin i.e. $\ell \in g$, their associated weights as displayed in (9) sum to one $\sum_{\ell \in g} \sum_e \omega_{e,\ell}^g = 1$.
- For relative periods belonging to some other bin included in (4) i.e. $\ell \in g'$ for $g' \neq g$ and $g' \in \mathcal{G}$, their associated weights as displayed in (10) sum to zero $\sum_{\ell \in g'} \sum_e \omega_{e,\ell}^g = 0$ for each bin g' .
- For relative periods not included in $\mathcal{G}$, their associated weights as displayed in (11) sum to negative one $\sum_{\ell \in g^{excl}} \sum_e \omega_{e,\ell}^g = -1$.
- If there are never-treated units i.e. $\infty \in \text{supp}(E_i)$, we have $\omega_{\infty,\ell}^g = 0$ for all g and ℓ .

We can easily estimate the weights $\omega_{e,\ell}^g$ for any given specification of $\mathcal{G}$ using the following auxiliary regression:

$$D_{i,t}^\ell \cdot \mathbf{1}\{E_i = e\} = \alpha_i + \lambda_t + \sum_{g \in \mathcal{G}} \omega_{e,\ell}^g \mathbf{1}\{t - E_i \in g\} + v_{i,t} \quad (12)$$

which regresses $D_{i,t}^\ell \cdot \mathbf{1}\{E_i = e\}$ on all bin indicators included in regression (4) and two-way fixed effects.

All of the above properties can be extended to a case where covariates are added to regression (4) by partialling out the covariates before proceeding. In other words, the terms in parentheses in (9), (10) and (11) would be replaced by terms for which the covariates are partialled out. The weights can be estimated by controlling for covariates in regression (12) the same way as they are controlled for in the original regression.

However, covariates complicate the interpretations of μ_g in terms of $CATT_{e,\ell}$ as we describe below in Proposition 2-4. Depending on how covariates are controlled for in regression (4), we may need an additional assumption that the counterfactual trends are linear in the time-varying covariates $X_{i,t}$. We leave a full investigation of the introduction of covariates for future work.

3.3 Interpreting the coefficients under parallel trends assumption only

Proposition 2. *Under Assumption 1 (parallel trends) only, the population regression coefficient on the indicator for relative period bin g is a linear combination of $CATT_{e,\ell \in g}$ as well as $CATT_{e,\ell'}$ from other*

relative periods $\ell' \notin g$, with the same weights stated in Proposition 1:

$$\mu_g = \sum_{\ell \in g} \sum_e \omega_{e,\ell}^g CATT_{e,\ell} + \sum_{g' \neq g, g' \in \mathcal{G}} \sum_{\ell' \in g'} \sum_e \omega_{e,\ell'}^g CATT_{e,\ell'} + \sum_{\ell' \in g^{excl}} \sum_e \omega_{e,\ell'}^g CATT_{e,\ell'}. \quad (13)$$

Under Assumption 1, the terms in Proposition 1 reduce to a linear combination of the causally interpretable building blocks $CATT_{e,\ell}$ as follows:

$$E[Y_{i,e+\ell} - Y_{i,0}^\infty | E_i] - E[Y_{i,e+\ell}^\infty - Y_{i,0}^\infty] = CATT_{e,\ell} + \underbrace{E[Y_{i,t}^\infty - Y_{i,0}^\infty | E_i] - E[Y_{i,t}^\infty - Y_{i,0}^\infty]}_{=0} \quad (14)$$

for $t = e + \ell$. However, two issues for interpretability remain. First, the coefficient μ_g can be written as an average of not only $CATT_{e,\ell}$ from own periods $\ell \in g$, but also $CATT_{e,\ell'}$ from other periods. Second, the weights are still non-linear functions of the distribution of cohorts, same as those in (12), and they are not restricted to lie in $[0, 1]$.

The properties of these weights as described following Proposition 1 imply that contamination from other periods wanes once we impose restrictions on treatment effects. In the next two subsections we illustrate how that can happen.

3.4 Interpreting the coefficients under parallel trends and no anticipation assumptions

Proposition 3. *If Assumption 1 (parallel trends) holds and Assumption 2 (no anticipatory behavior in all periods before the initial treatment) holds, the population regression coefficient μ_g is a linear combination of post-treatment $CATT_{e,\ell'}$ for all $\ell' \geq 0$, with the same weights stated in Proposition 1:*

$$\mu_g = \sum_{\ell' \in g, \ell' > 0} \sum_e \omega_{e,\ell'}^g CATT_{e,\ell'} + \sum_{g' \neq g, g' \in \mathcal{G}} \sum_{\ell' \in g', \ell' > 0} \sum_e \omega_{e,\ell'}^g CATT_{e,\ell'} + \sum_{\ell' \in g^{excl}, \ell' > 0} \sum_e \omega_{e,\ell'}^g CATT_{e,\ell'}. \quad (15)$$

Once we restrict pre-treatment $CATT_{e,\ell \leq 0}$ to be zero under the no anticipatory behavior assumption, the expression for μ_g simplifies as terms involving $CATT_{e,\ell \leq 0}$ drop out. However, the second term in the expression for μ_g remains unless we further impose treatment effect homogeneity for its summands to cancel out each other. Thus, μ_g may be non-zero for pre-treatment periods even if parallel trends holds.

This result immediately implies a shortcoming of using pre-treatment coefficients (i.e. μ_g where g contains only leads to the treatment $\ell < 0$) to test for pretrends. Under the no anticipatory behavior assumption,

The weight $\omega_\ell^g = \sum_e \omega_{e,\ell}^g$ sums over the weights $\omega_{e,\ell}^g$ from Proposition 1, and is equal to the population regression coefficient from the following auxiliary regression:

$$D_{i,t}^\ell = \alpha_i + \lambda_t + \sum_{g \in \mathcal{G}} \omega_{e,\ell}^g \mathbf{1}\{t - E_i \in g\} + v_{i,t} \quad (17)$$

which regresses $D_{i,t}^\ell$ on all bin indicators included in regression (4) and two-way fixed effects.

We note that even under treatment effect homogeneity μ_ℓ can still be contaminated by treatment effects from the excluded periods. This contamination, however, can be avoided by adjusting the specification to only exclude periods with zero treatment effect.

For specifications with relative time bins, we note that there can still be contamination from other bins as suggested by the second term of expression (16). A sufficient condition to avoid such contamination would be to group relative periods ℓ' into a bin only when their effects are the same since their weights $\omega_{\ell'}^g$ would sum to zero.

For the “fully dynamic” specification (7) where all g ’s are singletons of relative time periods, the weight $\omega_{\ell'}^\ell$ is zero for each relative period $\ell' \neq \ell$ that is included in the specification. The decomposition therefore simplifies to

$$\mu_\ell = ATT_\ell + \sum_{\ell' \in g^{excl}} \omega_{\ell'}^g ATT_{\ell'}. \quad (18)$$

3.6 Intuition for contamination

Proposition 3 demonstrates that even under the assumptions of parallel trends and no anticipation, estimates μ_g can still be contaminated by treatment effects from other periods. In this section we explain the intuition behind why this contamination occurs for the “fully dynamic” specification (7). A decomposition of μ_ℓ into a weighted average of $CATT_{e,\ell'}$ demonstrates that contamination is driven by the interaction of two elements: the weights and $CATT_{e,\ell'}$. The weights underlying the contamination are non-linear functions of the distribution of the cohorts. We do not attempt to provide a heuristic for determining the magnitude of the weights, but instead describe how to estimate the weights and later on in Section 4 how to estimate each $CATT_{e,\ell}$. This allows researchers to directly determine the degree of contamination in their application. Our publicly-available Stata package `eventstudyweights` automates the estimation of these weights using the panel dataset underlying any given specification of (1).

We apply the familiar omitted variable bias (OVB) formula to arrive at our decomposition. In an event study where individuals receive the treatment at different times, the panel can never be balanced in both calendar time and time relative to the initial treatment. As a result, the relative time indicators are still correlated even after controlling for unit and time fixed effects in a two-way fixed effects regression. We use the saturated regression and the OVB formula to illustrate how this correlation leads to contamination. We defer its formal derivation to Appendix B.

Under the parallel trends assumption only, the saturated regression is

$$\begin{aligned}
Y_{i,t} = & \sum_e E[Y_{i,0}^\infty | E_i = e] \cdot \mathbf{1}\{E_i = e\} + \sum_s E[Y_{i,s}^\infty - Y_{i,0}^\infty] \cdot \mathbf{1}\{t = s\} \\
& + \sum_{\ell' \in g^{incl}} \sum_{e \in I_{\ell'}} CATT_{e,\ell'} \cdot \left(D_{i,t}^{\ell'} \cdot \mathbf{1}\{E_i = e\} \right) \\
& + \sum_{\ell' \in g^{excl}} \sum_{e \in I_{\ell'}} CATT_{e,\ell'} \cdot \left(D_{i,t}^{\ell'} \cdot \mathbf{1}\{E_i = e\} \right) + \epsilon_{i,t}
\end{aligned} \tag{19}$$

where the regressors are cohort fixed effects, time fixed effects, and cohort-specific relative time indicators. Furthermore, let g^{incl} collect the relative time included in (7). The coefficient associated with the cohort-specific relative time indicator $D_{i,t}^\ell \cdot \mathbf{1}\{E_i = e\}$ is the cohort-specific average treatment effects $CATT_{e,\ell}$. To decompose the coefficient μ_ℓ from (7) in terms of this saturated regression (19), the OVB formula multiplies the coefficients in the saturated regression (19), $CATT_{e,\ell'}$, with the regression coefficients from (12), $\omega_{e,\ell'}^\ell$, which leads to the following decomposition for μ_ℓ as a linear combination of $CATT_{e,\ell'}$:

$$\mu_\ell = \sum_{e,\ell'} \omega_{e,\ell'}^\ell CATT_{e,\ell'}. \tag{20}$$

Since $\omega_{e,\ell'}^\ell$ is equal to a regression coefficient from (12), we can write it as

$$\omega_{e,\ell'}^\ell = (\sigma_{e,\cdot})^\top \mathbf{\Delta}_{e+\ell'} A_\ell^{-1}. \tag{21}$$

Below we briefly comment on each of the three elements in the above expression to highlight how they depend on the distribution of the cohorts. We defer their detailed definitions and derivations to Appendix A.1.

- $\sigma_{e,\cdot}$ is a vector of the covariance between cohort e and the other cohorts, namely $Cov(\mathbf{1}\{E_i = e\}, \mathbf{1}\{E_i = e'\})$.

This term thus scales quadratically in the share of cohort e , and is small for small cohorts.

- Δ_t is a matrix of demeaned relative time indicators. The entry that corresponds to cohort e' and relative time indicator $D_{i,t}^\ell$ is $E[D_{i,t}^\ell | E_i = e'] - \frac{1}{T+1} \mathbf{1}\{e' \in \mathcal{I}_\ell\}$. When T is large, i.e. the panel is long, the second term is small and this entry is therefore approximately equal to the relative time indicator.
- A_ℓ^{-1} is the row of A^{-1} that corresponds to the relative time indicator $D_{i,t}^\ell$ for A the covariance matrix of demeaned relative time indicators. Specifically, the entry in A that corresponds to the covariance between demeaned $D_{i,t}^\ell$ and $D_{i,t}^{\ell'}$ is

$$\sum_t Cov(D_{i,t}^\ell, D_{i,t}^{\ell'}) - \frac{1}{T+1} Cov(\mathbf{1}\{E_i \in \mathcal{I}_\ell\}, \mathbf{1}\{E_i \in \mathcal{I}_{\ell'}\}). \quad (22)$$

Within any time period $D_{i,t}^\ell$ and $D_{i,t}^{\ell'}$ are negatively correlated because no cohorts can be in these two relative times at the same time. The second covariance term is in general also non-zero because being in one cohort predicts (not) being in another cohort. Therefore A is in general not a diagonal matrix and A^{-1} would depend on the distribution of the cohorts non-linearly.

The three elements of (21) demonstrate the weights are non-linear functions of the distribution of the cohorts, and they are in general non-zero. Nonetheless these weights can be estimated easily by the auxiliary regression (12).

3.7 Invalidity of pretrend tests based on pre-period coefficients.

Contamination undermines the practice of testing for pretrends using pre-period coefficients. Proposition 3 implies that when effects are not homogenous across cohorts, it is problematic to interpret non-zero estimates for μ_g as evidence for pretrends, where the set g contains some leads $\ell < 0$. Proposition 4 implies that even with homogeneous treatment effect, if the effects associated with the excluded periods are not zero, then contamination may still occur. Therefore without strong assumptions, pre-period coefficients should not be used to test for pretrends because contamination can lead to estimates that are non-zero in the absence of pretrends or zero in the presence of pre-trends.

Testing for pretrends using pre-period coefficients is commonly used in practice. As an example, He and Wang (2017) mention “the estimated coefficients of the leads of treatments, i.e. δ_k for all $k \leq -2$